# IEDA 4000H

## Tutorial 2

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1. Credit Card Payment Problem
2. Tutorial on Norms
3. Robust Counterpart

# Problem Statement

Suppose that you have a credit card with an annual percentage rate of $12r$ (that is, a monthly periodic rate of $r$) and an initial balance of $\$B_0$. You decide to pay $\$p$ at the end of each month. Assume that no new purchases are made.

1. What is the balance on the credit card after one month?
2. What is the balance on the credit card after $n$ months?
3. If you want to pay off the entire balance within exactly $N$ months, what should the monthly payment be?

# Solution

1. After one month, the balance is

$$B_1 = B_0(1+r) - p.$$

2. After $n$ months, the balance is

$$B_n = B_{n-1}(1+r) - p,$$

which by recursion gives

$$B_n = B_0(1+r)^n - p\frac{(1+r)^n - 1}{r}.$$

## Solution

3 If you decide to pay off the entire balance after $N$ months, then

$$B_N = 0,$$

which implies

$$p = B_0 r \frac{(1+r)^N}{(1+r)^N - 1}.$$

# Introduction

Norms are fundamental concepts in linear algebra and functional analysis that provide a way to measure the "size" or "length" of vectors. They generalize the notion of Euclidean distance and are essential in various applications including optimization, numerical analysis, and machine learning.

## Definition of a Norm

A norm on a vector space $V$ over a field $\mathbb{F}$ (typically $\mathbb{R}$ or $\mathbb{C}$) is a function $\|\cdot\| : V \to [0, \infty)$ that satisfies the following properties for all vectors $\boldsymbol{u}, \boldsymbol{v} \in V$ and all scalars $a \in \mathbb{F}$:

1. Non-negativity: $\|\boldsymbol{v}\| \geq 0$ and $\|\boldsymbol{v}\| = 0$ if and only if $\boldsymbol{v} = 0$.
2. Scalar multiplication: $\|a\boldsymbol{v}\| = |a|\|\boldsymbol{v}\|$.
3. Triangle inequality: $\|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$.

## Common Vector Norms

- $l^1$-norm (Manhattan Norm): The $l^1$-norm of a vector $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)$ is defined as

$$\|\boldsymbol{v}\|_1 = |v_1| + |v_2| + \ldots + |v_n|.$$

- $l^2$-norm (Euclidean Norm): The $l^2$-norm of a vector $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)$ is defined as

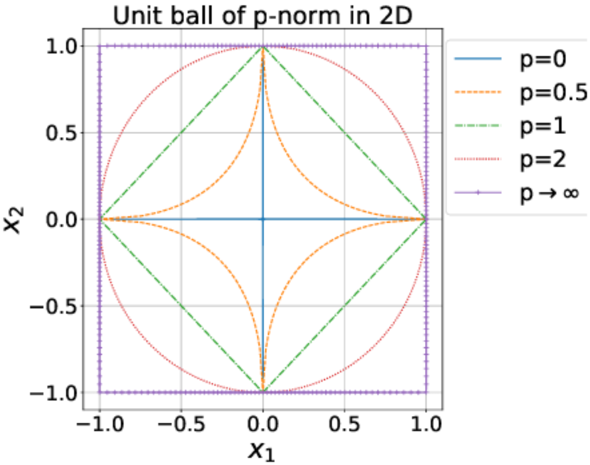
$$\|\boldsymbol{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \ldots + v_n^2}.$$

- $l^\infty$-norm (Maximum Norm): The $l^\infty$-norm of a vector $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)$ is defined as

$$\|\boldsymbol{v}\|_\infty = \max(|v_1|, |v_2|, \ldots, |v_n|).$$

- General $l^p$-norm: For $1 \le p < \infty$, the $l^p$-norm of a vector $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)$ is defined as

$$\|\boldsymbol{v}\|_p = \left(|v_1|^p + |v_2|^p + \ldots + |v_n|^p\right)^{1/p}.$$


Unit ball of p-norm in 2D
1.0
0.5
0.0
−0.5
−1.0
x2
−1.0
−0.5
0.0
0.5
1.0
x1
p=0
p=0.5
p=1
p=2
p → ∞

# Matrix Norms

Matrix norms extend the concept of vector norms to matrices. A matrix norm $\|\cdot\|$ on the space of $m \times n$ matrices must satisfy similar properties as vector norms.

- Frobenius Norm: The Frobenius norm of a matrix $\mathrm{A} = [a_{ij}]$ is defined as

$$\|\mathrm{A}\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}|^2}.$$

- Induced Norms: For a given vector norm $\|\cdot\|$, the induced matrix norm is defined as

$$\|\mathrm{A}\| = \sup_{\boldsymbol{x}\neq 0} \frac{\|\mathrm{A}\boldsymbol{x}\|}{\|\boldsymbol{x}\|}.$$

  Common induced norms include:
  - $l^1$-induced norm: Maximum absolute column sum.
  - $l^\infty$-induced norm: Maximum absolute row sum.
  - $l^2$-induced norm: Largest singular value of the matrix.

## Properties of Norms

- Equivalence of Norms: In finite-dimensional vector spaces, all norms are equivalent, meaning they induce the same topology. However, they may differ in terms of numerical values and computational properties.
- Cauchy-Schwarz Inequality: For any vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ in an inner product space, the following inequality holds:

$$|\langle \boldsymbol{u}, \boldsymbol{v} \rangle| \leq \|\boldsymbol{u}\|_2 \|\boldsymbol{v}\|_2.$$

In general, for $\frac{1}{p} + \frac{1}{q} = 1,\ 0 < p, q < \infty$, we have

$$|\langle \boldsymbol{u}, \boldsymbol{v} \rangle| \leq \|\boldsymbol{u}\|_p \|\boldsymbol{v}\|_q.$$

Here, $\|\cdot\|_p$ and $\|\cdot\|_q$ are dual norms. In particular, the dual norm of $\|\cdot\|_1$ is $\|\cdot\|_\infty$.

# Exercises

1. Show that for any vector $\boldsymbol{v} \in \mathbb{R}^n$, the following inequalities hold:

$$\|\boldsymbol{v}\|_\infty \leq \|\boldsymbol{v}\|_1 \leq n\|\boldsymbol{v}\|_\infty.$$

2. Show that for any vector $\boldsymbol{v} \in \mathbb{R}^n$, the following inequalities hold:

$$\|\boldsymbol{v}\|_\infty \leq \|\boldsymbol{v}\|_2 \leq \sqrt{n}\|\boldsymbol{v}\|_\infty.$$

## Solutions to Exercises

1 Proof:

- First, we show that $\|\boldsymbol{v}\|_\infty \leq \|\boldsymbol{v}\|_1$:

$$\|\boldsymbol{v}\|_\infty = \max(|v_1|, |v_2|, \ldots, |v_n|) \leq |v_1| + |v_2| + \ldots + |v_n| = \|\boldsymbol{v}\|_1.$$

- Next, we show that $\|\boldsymbol{v}\|_1 \leq n\|\boldsymbol{v}\|_\infty$:

$$\|\boldsymbol{v}\|_1 = |v_1| + |v_2| + \ldots + |v_n| \leq n\max(|v_1|, |v_2|, \ldots, |v_n|) = n\|\boldsymbol{v}\|_\infty.$$

2 Proof:

- First, we show that $\|\boldsymbol{v}\|_\infty \leq \|\boldsymbol{v}\|_2$:

$$\|\boldsymbol{v}\|_\infty = \max(|v_1|, |v_2|, \ldots, |v_n|) \leq \sqrt{v_1^2 + v_2^2 + \ldots + v_n^2} = \|\boldsymbol{v}\|_2.$$

- Next, we show that $\|\boldsymbol{v}\|_2 \leq \sqrt{n}\|\boldsymbol{v}\|_\infty$:

$$\|\boldsymbol{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \ldots + v_n^2} \leq \sqrt{n}\max(|v_1|, |v_2|, \ldots, |v_n|) = \sqrt{n}\|\boldsymbol{v}\|_\infty.$$

# Robust Counterpart for Ellipsoidal Uncertainty Set

Consider the linear robust constraint

$$(\bar{\boldsymbol{a}} + \mathrm{P}\boldsymbol{z})^\top \boldsymbol{x} \le b, \quad \forall \boldsymbol{z} \in \mathcal{Z},$$

where $\mathcal{Z} = \{\boldsymbol{z} : \|\boldsymbol{z}\|_2 \le \rho\}$ is an ellipsoidal uncertainty set. Show that this robust constraint is equivalent to the second-order cone constraint

$$\bar{\boldsymbol{a}}^\top \boldsymbol{x} + \rho \|\mathrm{P}^\top \boldsymbol{x}\|_2 \le b.$$

Proof: The robust constraint can be rewritten as

$$\bar{\boldsymbol{a}}^\top \boldsymbol{x} + \max_{\|\boldsymbol{z}\|_2 \le \rho} (\mathrm{P}\boldsymbol{z})^\top \boldsymbol{x} \le b.$$

By the Cauchy-Schwarz inequality, we have

$$(\mathrm{P}\boldsymbol{z})^\top \boldsymbol{x} \le \|\boldsymbol{z}\|_2 \|\mathrm{P}^\top \boldsymbol{x}\|_2 \le \rho \|\mathrm{P}^\top \boldsymbol{x}\|_2.$$

Therefore, the robust counterpart for ellipsoidal uncertainty set is

$$\bar{\boldsymbol{a}}^\top \boldsymbol{x} + \rho \|\mathrm{P}^\top \boldsymbol{x}\|_2 \le b.$$

# Robust Counterpart for 1-norm Uncertainty Set

Consider the linear robust constraint

$$(\overline{\boldsymbol{a}} + \mathrm{P}\boldsymbol{z})^\top \leq b, \quad \forall \boldsymbol{z} \in \mathcal{Z},$$

where $\mathcal{Z} = \{\boldsymbol{z} : \|\boldsymbol{z}\|_1 \leq \rho\}$ is a 1-norm uncertainty set. Show that this robust constraint is equivalent to the following constraint

$$\overline{\boldsymbol{a}}^\top \boldsymbol{x} + \rho \|\mathrm{P}\boldsymbol{x}\|_\infty \leq b.$$

Proof: The robust constraint can be rewritten as

$$\overline{\boldsymbol{a}}^\top \boldsymbol{x} + \max_{\|\boldsymbol{z}\|_1 \le \rho} (\mathrm{P}\boldsymbol{z})^\top \boldsymbol{x} \le b.$$

By the Cauchy-Schwarz inequality, we have

$$(\mathrm{P}\boldsymbol{z})^\top \boldsymbol{x} \le \|\boldsymbol{z}\|_1 \|\mathrm{P}^\top \boldsymbol{x}\|_\infty \le \rho \|\mathrm{P}^\top \boldsymbol{x}\|_\infty.$$

Therefore, the robust counterpart for 1-norm uncertainty set is

$$\overline{\boldsymbol{a}}^\top \boldsymbol{x} + \rho \|\mathrm{P}^\top \boldsymbol{x}\|_\infty \le b.$$

Thank you for listening !

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